The Minimal additive Model

Roberto Baviera[‡]

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(‡) Politecnico di Milano, Department of Mathematics, 32 p.zza L. da Vinci, Milano

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Abstract

We (briefly) describe the most elementary additive process for derivative pricing. The model can be considered a toy model, but able to reproduce some stylized facts observed in market implied volatility surfaces within a coherent model. An explicit CDF of the increment between any future dates enables for an extremely fast Monte Carlo simulation of any discretely-monitored derivative for pricing and risk management.

Keywords: volatility surface, Bachelier model, additive process, cascade calibration.

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Address for correspondence:
Roberto Baviera
Department of Mathematics
Politecnico di Milano
32 p.zza Leonardo da Vinci
I-20133 Milano, Italy
Tel. +39-02-2399 4575
roberto.baviera@polimi.it

The Minimal Additive model

1 The model

Let us consider $F(t,T)$ a forward valued at time t and expiry at time T.
additive processes allow a simple description for both linear models (or Bachelier-like, where the underling can assume any real value) and exponential processes (or Black-like, where the underling can assume only positive values).
In the linear additive model, the forward is modeled as

$$F(t,T) = F(0,T) + f_t \quad \forall t \in [0,T] \ , \tag{1}$$

where $\{f_t\}_{t\geq 0}$ is an additive process (then with independent increments).
In the exponential additive model, the forward is

$$F(t,T) = F(0,T) e^{f_t} \quad \forall t \in [0,T] \ , \tag{2}$$

where $\{f_t\}_{t\geq 0}$ is an additive process. In both cases the forward $\{F(t,T)\}_{t\geq 0}$ is a martingale process.

1.1 The Minimal additive model

We describe $\{f_t\}_{t\geq 0}$ with an additive process

$$f_t = \sigma_t \sqrt{t}\, \zeta_t$$

introducing the auxiliary process $\{\zeta_t\}$ where we have removed the scale.
The process is based on an *asymmetric Laplace distribution* (Kozubowski and Podgórski 1999)[1], that describes the marginal of ζ_t at every time t, a family of infinite divisible distributions. The probability density function of ζ at time t is

$$\mathscr{P}_t(\zeta) = C_t \left(e^{\alpha_t\,(\zeta-\gamma_t)}\ \mathbb{1}_{\zeta<\gamma_t} + e^{-\beta_t\,(\zeta-\gamma_t)}\ \mathbb{1}_{\zeta\geq\gamma_t} \right) \ , \tag{3}$$

where $\alpha_t, \beta_t > 0$, C_t is the normalization constant and γ_t is the localization parameter; It is used to impose the martingality condition for the forward price $F(t,T)$. The martingality condition implies that, for linear models $\mathbb{E}_0[f_t] = 0 \Rightarrow \mathbb{E}_0[\zeta_t] = 0$. For exponential model the condition becomes $\mathbb{E}_0[e^{f_t}] = 1$.
Model parameters are three, strictly positive continuous functions of time, all with a clear financial interpretation. The scale σ_t models the term structure of the volatility; α_t the left tail of the distribution and β_t the right tail.

1.2 Basic properties

The key quantities for ζ are

[1] Called also *two-sided exponential distribution* by Sato (1999, example 15.14, Ch.3, p.98).

Quantity	Symbol	Description
pdf	$\mathscr{P}_t(\zeta)$	$C_t\, e^{A_t(\zeta-\gamma_t)-B_t\lvert\zeta-\gamma_t\rvert}$
A_t		$\frac{1}{2}\,(\alpha_t-\beta_t)$
B_t		$\frac{1}{2}\,(\alpha_t+\beta_t)$
C_t		$\left(\frac{1}{\beta_t}+\frac{1}{\alpha_t}\right)^{-1}$
LCF	$\phi_t(u) := \ln \mathbb{E}[e^{i\,u\,\zeta}]$	$-\ln\left(1-i\,u\left(\frac{1}{\beta_t}-\frac{1}{\alpha_t}\right)+\frac{u^2}{\beta_t\,\alpha_t}\right)+i\gamma_t\,u$
mean	$\mathbb{E}[\zeta]$	$\gamma_t+\frac{1}{\beta_t}-\frac{1}{\alpha_t}$
variance	$\mathbb{V}\mathrm{ar}[\zeta]$	$\frac{1}{(\beta_t)^2}+\frac{1}{(\alpha_t)^2}$
skewness		$2\left(1-\left(\frac{\beta_t}{\alpha_t}\right)^3\right)/\left(1+\left(\frac{\beta_t}{\alpha_t}\right)^2\right)^{3/2}$
excess Kurtosis		$6\left(1+\left(\frac{\beta_t}{\alpha_t}\right)^4\right)/\left(1+\left(\frac{\beta_t}{\alpha_t}\right)^2\right)^2$

Let us notice that the distribution of ζ is self-decomposable and then infinite divisible. The process $\{f_t\}_{t\geq 0}$ is an infinite activity process. The LCF for f_t is analytic in the horizontal strip $(-p_t^+, p_t^-)$

Hint. The LCF for f_t should be considered for the pricing and for the martingale condition of the exponential model.

1.3 Parameters' constraints

In order to have an additive process that properly describes the dynamics of the forward price, model parameters should satisfy the following constraints:

parameter	constraint
$p_t^{\pm}$	decreasing
$\sigma_t\sqrt{t}$	increasing

In addition, the exponential model should satisfy the condition

$$p_t^+ > 1 \quad \forall t \in [0, T].$$

2 Call option price in linear and exponential models

Maturity T, strike K, discout $B_0 := B(0,T)$ and moneyness k.
European call option

$$C(T,K) = B_0\,\mathbb{E}[F(T,T)-K]^+$$

Closed formula for European call option in the linear case

$$c_b(T,K) = B_0\,\mathbb{E}[f_T-k]^+ = B_0\,\sigma_T\,\sqrt{T}\begin{cases} C_T\dfrac{e^{\alpha_T(\tilde{k}-\gamma_T)}}{(\alpha_T)^2}-\tilde{k} & \text{for } \tilde{k}<\gamma_T \\ C_T\dfrac{e^{-\beta_T(\tilde{k}-\gamma_T)}}{(\beta_T)^2} & \text{for } \tilde{k}\geq\gamma_T \end{cases}$$

where $\tilde{k} := \frac{k}{\sigma_T \sqrt{T}}$.
Closed formula for European call option in the exponential case (for $p_T^+ > 1$)

$$C(T,K) = B_0\, F(0,T)\, \mathbb{E}[e^{f_T} - e^k]^+$$

$$C(T,K) = B_0\, F(0,T) \begin{cases} \dfrac{C_T}{\sigma_T \sqrt{T}} \left\{ \dfrac{e^{k\,(p_T^- + 1) - p_T^- \mu_T}}{p_T^-\,(p_T^- + 1)} + \dfrac{e^{\mu_T}\,(p_T^- + p_T^+)}{(p_T^- + 1)(p_T^+ - 1)} \right\} - e^k & \text{for } k < \mu_T \\ \dfrac{C_T}{\sigma_T \sqrt{T}}\, \dfrac{e^{-k\,(p_T^+ - 1) + p_T^+ \mu_T}}{p_T^+\,(p_T^+ - 1)} & \text{for } k \geq \mu_T \end{cases}$$

Given the implied vol $I(k,T)$, the vol–skew is

$$\text{vol} - \text{skew} := \frac{\partial I(k,T)}{\partial k}|_{k=0}$$

Quantity	Linear (additive Bachelier)	Exponential (additive Black)
$F(t,T)$	$F(0,T)\, + f_t$	$F(0,T)\, e^{f_t}$
k	$K - F(0,T)$	$\ln \dfrac{K}{F(0,T)}$
γ_t	$-\dfrac{1}{\beta_t} + \dfrac{1}{\alpha_t}$	???
vol–skew	$\begin{cases} \sqrt{2\pi}\left(\dfrac{1}{2} - \dfrac{e^{-1+\frac{\beta_t}{\alpha_t}}}{1+\frac{\beta_t}{\alpha_t}}\right) & \gamma_t \leq 0 \\ -\sqrt{2\pi}\left(\dfrac{1}{2} - \dfrac{e^{-1+\frac{\alpha_t}{\beta_t}}}{1+\frac{\alpha_t}{\beta_t}}\right) & \gamma_t \geq 0 \end{cases}$	???

Hint: Check for possible errors in the formulas.

3 Benchmark models

The benchmark models we consider are

Linear (additive Bachelier)	Exponential (additive Black)
additive Bachelier (Baviera and Massaria 2026)	ATS (Azzone and Baviera 2022)
Generalized Logistic (GL)	GBA (Azzone and Torricelli 2025)

For linear models, the first benchmark model is the additive Bachelier (Baviera and Massaria 2026), in particular consider it with α chosen equal to $1/2$.
The second benchmark model considers the Type IV Generalized Logistic (GLIV) distribution to model the pdf of ζ_t, a simple ratio of exponential functions. GLIV distributions has been introduced by Prentice (1976) and whose properties have been prominently studied under the name of z-distributions (ZD) class in Barndorff-Nielsen *et al.* (1982)

$$\mathscr{P}_t(\zeta) = \frac{\Gamma(\alpha_t + \beta_t)}{\Gamma(\alpha_t)\,\Gamma(\beta_t)}\, \frac{e^{\alpha_t(\zeta - \gamma_t)}}{\left(1 + e^{(\zeta - \gamma_t)}\right)^{\alpha_t + \beta_t}}$$

with $\Gamma(\bullet)$ the Gamma function. In the following, we call Generalized Logistic (GL) the model that uses a GLIV as pdf of ζ_t for a given t: it a generalization of the model in Carr and Torricelli (2021, Sec.3), whose exponential equivalent is the GBA in Azzone and Torricelli (2025).

For exponential models, the first benchmark model is the ATS (Azzone and Baviera 2022), in particular consider it with α chosen equal to 1/2.
The other benchmark is GBA (Azzone and Torricelli 2025). The GBA is an additive process for exponential models, that allows a closed form solution for European call options also in the real space (Azzone and Torricelli 2025). The process is based on the GLIV distribution and the exponential GLIV, called generalized beta in Azzone and Torricelli (2025). It is characterized by three continuous deterministic functions $\alpha(t)$, $\beta(t)$ and $\sigma(t)$ that should satisfy some inequalities (cf., e.g., Azzone and Torricelli 2025, Table 5, p.774) in the calibration procedure. GBA main advantage is that - beyond sharing all nice features of other additive processes - it has designed for the simplicity in the closed form solution that involves an upper incomplete Beta functions that, thanks to the relation with the Gauss hypergeometric function, allows a series representation.

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Notation and shorthands

Symbol	**Description**
$\mathbb{E}_s[\bullet]$	$\mathbb{E}_s[\bullet \mid \mathscr{F}_s]$ where $\mathscr{F}_s$ is the natural filtration
B_0	discount factor between the value date and option maturity T
$F(t,T)$	Forward price with expiry in T and valued at time $t \in [0,T]$
$C_b\,(k,T;\sigma)$	Bachelier call option price wrt moneyness k and maturity T, with volatility σ
$C\,(k,T;\mathbf{p})$	Model call option price wrt moneyness and maturity, function of parameters $\mathbf{p}$

Some other quantities

Description	Quantity	Rescaled Quantity
	f_t	ζ_t
option moneyness	k	$\tilde{k} := \dfrac{k}{\sigma_t \sqrt{T}}$
left pdf tail	$p_t^- = \dfrac{\alpha_t}{\sigma_t \sqrt{T}}$	α_t
right pdf tail	$p_t^+ = \dfrac{\beta_t}{\sigma_t \sqrt{T}}$	β_t
localization	$\mu_t := \gamma_t \, \sigma_t \sqrt{T}$	γ_t

Symbol	Description
ATM	At-The-Money (referred to options)
bp	basis point (typical measurement unit for rates)
CDF	cumulative distribution function
OTM	Out-The-Money (referred to options)
pdf	probability density function
rv	random variable
wrt	with respect to